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May 30, 1975
SUPERSEDING
Fed. Spec. UU-T-450D
June 5, 1969

# FEDERAL SPECIFICATION

# TISSUE, FACIAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and two sizes of a soft, highly absorbent, two-ply, folded facial tissue.

1.2 Classification. Facial tissue shall be furnished in the following sizes, as specified (see 6.2).

Size 1 - Large.
Size 2 - Small.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Test Methods:

T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 432 - Water Absorption of Bibulous Papers.
T 435 - Hydrogen Ion Concentration (pH) of Paper Extracts - Hot Extraction Method.
T 452 - Brightness of Paper and Paperboard.
T 457 - Stretch of Paper and Paperboard.
T 498 - Softness of Sanitary Tissues.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Facial tissues shall be made from fibrous, cellulosic material which may include reclaimed fibers.

3.2 Construction. The facial tissues shall consist of two plies of tissue material suitably combined to form the end item tissue. The two-ply tissues shall conform to the requirements of this specification. The tissues shall be assembled to dispense one tissue at a time without disturbing the underlying tissue. When specified (see 6.2), the tissues shall be so assembled that when dispensed from a carton, removal of one tissue dispenses a portion of the next tissue ("pop-up" style).

3.3 Color. The color of the tissues shall be white.

3.4 Size 1. The area of each single sheet of size 1 tissue shall be not less than 80 square inches with the shortest dimension not less than 8 inches.

3.5 Size 2. The area of each single sheet of size 2 tissues shall be not less than 40 square inches with the shortest dimension not less than 5 inches.

3.6 Physical and chemical properties. The end item tissue shall conform to the requirements of table I when tested as specified in 4.3.2.



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TABLE I. Physical and chemical properties

| Characteristic | |
|---|---|
| Basis weight, 24 by 36 inches, 500 two-ply sheets, not less than, pounds | 19 |
| Tensile strength, average, both directions, not less than, kilograms per 15 millimeters of width. | 2.1 |
| Stretch, machine direction, not less than, percent | 8 |
| Softness value, total, both directions not more than, grams | 24 |
| Brightness, not less than, percent | 76 |
| Absorption, 0.01 milliliter of water, maximum, seconds | 30 |
| Acidity, pH value, hot extraction, not less than | 4.5 |

3.7 Workmanship. Facial tissue shall be pliable, soft with uniform formation and free from holes, lumps, lint, shives, slivers, dirt, abrasive particles, or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling for inspection and test. Sampling for inspection and test shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For sampling purposes, a lot shall consist of all tissues offered for inspection at one time.

4.2 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be made for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth therein.

4.3.1.1 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with requirements of section 5. The lot size shall be expressed in units of shipping containers and shall be the number of containers offered for inspection at one time. The sample unit shall be one container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of defects per hundred units. Defects shall be scored as specified below.

| Examine | Defects |
|---|---|
| Contents | Not as specified. |
| Containers | Not as specified. |
| Markings | Omitted; incorrect; illegible; improper size, location, sequence or method of application. |
| Materials | Component missing or damaged. |
| Workmanship | Bulging or distortion of container. Cushioning inadequate, improper or missing. |



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4.3.1.1.1 Examination of closure, waterproofing and banding of shipping containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix of that specification.

4.3.1.1.2 Examination of the immediate carton. An examination shall be made for the defects set forth below. The lot size shall be expressed in units of cartons comprising the inspection lot. The sample unit shall be one carton. The inspection level shall be S-3 and the AQL 2.5, expressed as percent defective. The cartons comprising the sample shall be selected at random from the shipping containers selected in 4.3.1.1.

| Examine | Defects |
|---|---|
| Appearance | Not dispensing type, as applicable; does not dispense as specified. Cartons not sealed tightly. |
| Workmanship | Cartons dirty, or torn. Holes in carton walls. |

4.3.1.1.3 Examination of the item for count. Two cartons shall be selected at random from each of the shipping containers selected in 4.3.1.1. The lot shall be unacceptable if the average count of tissues per carton, of all cartons examined, is less than specified.

4.3.1.2 Examination of the end item for visual and dimensional defects. Examine 100 sheets of facial tissue for the defects shown below. The 100 sheets shall be accumulated by selecting an approximate equal number of sheets from each of the cartons selected in accordance with 4.3.1.1.2. The lot shall be unacceptable if more than 2 of the 100 sheets examined contain one or more of the defects listed below:

| Examine | Defects |
|---|---|
| Appearance | Not folded. Not 2-ply. |
| Size | Area less than 80 square inches; shortest dimension less than 8 inches for size 1. Area less than 40 square inches; shortest dimension less than 5 inches for size 2. |
| Workmanship | Soiled, presence of holes, dirt spots, lint, lumps, shives, slivers, abrasive particles, or foreign matter. Formation not uniform. |

4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics in table II. The method of testing as listed in table II shall be followed. The lot size shall be expressed in units of cartons of the end item tissue. The test sample unit shall consist of 25 sheets of 2-ply facial tissue selected at random from the cartons selected under 4.3.1.1.2 and not more than 5 sheets shall be taken from any one carton. The inspection level for determining the sample size shall be S-1. Test results shall include all values upon which the results are based and there shall be no failures to meet the lot average or composite requirements. The composite shall be made up by combining approximately equal portions of small pieces of tissue from all sample units selected. All samples shall be conditioned and tested in accordance with the provisions of TAPPI method T 402.



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TABLE II. Testing of the end item

| Characteristic | Requirement paragraph | Test method 1/ | Results applicable to | No. determinations per sample unit | Results reported numerically to nearest |
|---|---|---|---|---|---|
| Atmospheric conditions | 4.3.2 | T 402 | All testing | | |
| Basis weight | Table I | T 410 | Lot average | 1 | 0.1 pound |
| Tensile strength | Table I | T 404 | Lot average | 10 each direction | 0.1 gms/15 mm |
| Stretch, machine direction | Table I | T 457 | Lot average | 10 | 0.1 percent |
| Softness value | | | | | |
| Machine dir. | Table I | T 498 | Lot average | 1 per side | 0.5 gram |
| Cross dir. | Table I | T 498 | Lot average | 1 per side | 0.5 gram |
| Brightness | Table I | T 452 | Lot average | 6 | 0.1 percent |
| Absorption 0.01 ml. water | Table I | T 432 | Lot average | 10 | 1 second |
| Adidity, pH | Table I | T 435 | Composite | 2 on composite | 0.1 pH |

1/ Refers to TAPPI method.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The folded two-ply facial tissues of one size only, assembled as specified in 3.2, in quantities specified in 6.2 and table III shall be packaged in a paperboard dispensing carton normally used in the industry. The carton shall be sealed to prevent accidental opening.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The number of cartons specified in table III shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed and banded in accordance with the appendix to the box specification.

TABLE III. Quantities per container

| Size | 2-ply tissues per paperboard carton | Cartons per shipping container |
|---|---|---|
| 1 | 100 | 36 |
| | 200 | 36 |
| 2 | 50 | 200 |

5.2.2 Level B. The number of cartons specified in table III shall be packed in a box conforming to PPP-B-636, class domestic, grade not less than 175. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. Facial tissues in quantities as specified (see 6.2), shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military activities, as applicable (see 6.2).



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6. NOTES

6.1 Intended use. Facial tissue covered by this specification is suitable for use as a disposable handkerchief, for oral and nasal sanguineous or infectious discharges, as nursing wipes, and other similar hospital and institutional uses.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) "Pop-up" style (see 3.2).
(d) Number of tissues per carton (see 5.1).
(e) Selection of levels of packaging and packing (see 5.1 and 5.2).
(f) Appropriate marking document (see 5.3).

The Department of Defense has waived coordination of this document.

MILITARY INTEREST:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.



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